

Mark Rober's CrunchLabs Backyard Squirrel Obstacle Course

Time	Subtitle	Human Translation
0s	This is a bird feeder,	This is a bird feeder,
1s	and everything to my left is my attempt at making it squirrel-proof.	and everything to my left is my attempt at making it squirrel-proof.
4s	If they want the birdseed, they need to pass through	If they want the birdseed, they need to pass through
7s	what is basically a Ninja Warrior obstacle course for squirrels.	what is basically a Ninja Warrior obstacle course for squirrels.
11s	This course is extremely challenging. It is not for the timid of heart.	This course is extremely challenging. It is not for the timid of heart.
16s	But out of the gate, I will admit that, in hindsight,	But out of the gate, I will admit that, in hindsight,
18s	that I completely underestimated my adversary.	that I completely underestimated my adversary.
21s	If you're wondering why I'd go through all this,	If you're wondering why I'd go through all this,
23s	we need to go back eight weeks ago	we need to go back eight weeks ago
25s	when I found myself stuck at home and very bored.	when I found myself stuck at home and very bored.
28s	So I installed a bird feeder and decided to become a bird-watcher.	So I installed a bird feeder and decided to become a bird-watcher.
32s	And it was just so lovely.	And it was just so lovely.
35s	Until they showed up.	Until they showed up.
37s	Luckily, my bird feeder had a wire cage around it	Luckily, my bird feeder had a wire cage around it
39s	big enough for the birds to slip inside,	big enough for the birds to slip inside,
42s	but small enough to prevent the squirrels from getting the food.	but small enough to prevent the squirrels from getting the food.
45s	And that worked really well until right here.	And that worked really well until right here.
48s	For me, this was like the moment in Jurassic Park	For me, this was like the moment in Jurassic Park
50s	when they realize the velociraptors can open door handles.	when they realize the velociraptors can open door handles.
53s	And once the door was open, it was clear this wasn't their first rodeo.	And once the door was open, it was clear this wasn't their first rodeo.
57s	They basically cleared this whole tube of birdseed by the end of the day.	They basically cleared this whole tube of birdseed by the end of the day.
1:01	This bird feeder was advertised as being squirrel-proof,	This bird feeder was advertised as being squirrel-proof,
1:04	so I bought another squirrel-proof bird feeder to replace it.	so I bought another squirrel-proof bird feeder to replace it.

1:07	It's got this outer cage suspended by some springs,	It's got this outer cage suspended by some springs,
1:10	so if a bird lands here, it doesn't weigh that much,	so if a bird lands here, it doesn't weigh that much,
1:13	and it has access to the seeds.	and it has access to the seeds.
1:15	However, if a squirrel comes, they'll grab onto this cage.	However, if a squirrel comes, they'll grab onto this cage.
1:18	Its weight will force the springs down,	Its weight will force the springs down,
1:20	thereby preventing access to the seeds.	thereby preventing access to the seeds.
1:23	I have no idea how the squirrels could possibly outsmart this,	I have no idea how the squirrels could possibly outsmart this,
1:27	but let's see.	but let's see.
1:28	And this one worked exactly as intended.	And this one worked exactly as intended.
1:31	The birds love it, but when the squirrel comes along,	The birds love it, but when the squirrel comes along,
1:33	the cage shifts down, and now all the seed ports are covered up.	the cage shifts down, and now all the seed ports are covered up.
1:37	Every time they tried, it was the same results.	Every time they tried, it was the same results.
1:40	The next morning,	The next morning,
1:41	this guy manages to unclip the springs that hold the cage up.	this guy manages to unclip the springs that hold the cage up.
1:44	The cage stays in the lowered position where it blocks all the seed openings.	The cage stays in the lowered position where it blocks all the seed openings.
1:48	Problem with that is now they can get to the lid.	Problem with that is now they can get to the lid.
1:51	But this bird feeder lid doesn't hinge open to give direct access.	But this bird feeder lid doesn't hinge open to give direct access.
1:55	So he goes with a different tactic...	So he goes with a different tactic...
2:00	...where he now has access to all the birdseed he wants.	...where he now has access to all the birdseed he wants.
2:03	And so I decided I would try one more bird feeder design.	And so I decided I would try one more bird feeder design.
2:06	Okay. Wow.	Okay. Wow.
2:09	It almost feels intentional at this point.	It almost feels intentional at this point.
2:11	This one's supposed to work well. It's got this baffle,	This one's supposed to work well. It's got this baffle,
2:13	which is supposed to keep the squirrels from getting to the seeds.	which is supposed to keep the squirrels from getting to the seeds.
2:17	And more importantly, there's no way for them to rip this off.	And more importantly, there's no way for them to rip this off.
2:20	We'll see.	We'll see.
2:21	This one debuted to similar results.	This one debuted to similar results.
2:24	As he hangs by a few toes,	As he hangs by a few toes,

2:26	I really started to gain an appreciation for what they were capable of.	I really started to gain an appreciation for what they were capable of.
2:30	It definitely seemed like this bird feeder was their favorite.	It definitely seemed like this bird feeder was their favorite.
2:33	Okay. Yeah, they're flexing on me for sure.	Okay. Yeah, they're flexing on me for sure.
2:36	And at this point, a plan was forming in my head,	And at this point, a plan was forming in my head,
2:38	but to be thorough, I moved the post away from the fence	but to be thorough, I moved the post away from the fence
2:41	and I can't say I was surprised when they climbed it.	and I can't say I was surprised when they climbed it.
2:43	Nor was I surprised	Nor was I surprised
2:45	to see they could shimmy up and down a small metal pole	to see they could shimmy up and down a small metal pole
2:48	or even a large one, for that matter.	or even a large one, for that matter.
2:50	Whether they could climb a certain pole material was a moot point	Whether they could climb a certain pole material was a moot point
2:54	because of this.	because of this.
2:55	Look at this! These guys are basically freaking rodent-sized Simone Biles.	Look at this! These guys are basically freaking rodent-sized Simone Biles.
3:00	This is what it felt like when someone stole a package from my porch,	This is what it felt like when someone stole a package from my porch,
3:04	and I harnessed that indignation,	and I harnessed that indignation,
3:06	spending a year designing the Glitter Bomb bait package...	spending a year designing the Glitter Bomb bait package...
3:09	Hello.	Hello.
3:12	...complete with special fragrances.	...complete with special fragrances.
3:14	[Mark] If I could design a package to deter thieves,	[Mark] If I could design a package to deter thieves,
3:17	could I design a bird feeder to deter squirrels?	could I design a bird feeder to deter squirrels?
3:21	After a couple of weeks of brainstorming, designing and building with my buddy John,	After a couple of weeks of brainstorming, designing and building with my buddy John,
3:25	we put it all together in this 20-second build montage.	we put it all together in this 20-second build montage.
3:45	Here are the basic elements.	Here are the basic elements.
3:47	Here's where I've placed their favorite bird feeder,	Here's where I've placed their favorite bird feeder,
3:50	at the end of the course.	at the end of the course.
3:51	And to sweeten the deal, right below that, there's a trapdoor.	And to sweeten the deal, right below that, there's a trapdoor.
3:55	And if they step here,	And if they step here,
3:57	it releases a butt-ton of walnuts	it releases a butt-ton of walnuts

3:59	and unfurls some celebratory banners.	and unfurls some celebratory banners.
4:02	Why walnuts? I'm glad you asked.	Why walnuts? I'm glad you asked.
4:04	Because over the course of a week,	Because over the course of a week,
4:06	I put out a buffet of seven different nuts and seeds.	I put out a buffet of seven different nuts and seeds.
4:09	All four times I repeated the experiment,	All four times I repeated the experiment,
4:11	walnuts were always the one they ate first.	walnuts were always the one they ate first.
4:13	What's great about placing this here is it levels the playing field,	What's great about placing this here is it levels the playing field,
4:17	as the birds can fly up and get as much food as they want anytime they need it.	as the birds can fly up and get as much food as they want anytime they need it.
4:21	The squirrels, however, will have to work for it	The squirrels, however, will have to work for it
4:23	because the only way to make it to this platform	because the only way to make it to this platform
4:26	is if they work their way through my Ninja Warrior obstacle course,	is if they work their way through my Ninja Warrior obstacle course,
4:29	which starts right here.	which starts right here.
4:31	This is the only platform on the course	This is the only platform on the course
4:33	that has a pole that isn't covered in slippery stuff,	that has a pole that isn't covered in slippery stuff,
4:35	which means this is where they have to climb up to enter the course.	which means this is where they have to climb up to enter the course.
4:39	If they try and get up any other way,	If they try and get up any other way,
4:40	they can't keep a good grip and they slide back down.	they can't keep a good grip and they slide back down.
4:43	The first challenge is the Bridge of Instability.	The first challenge is the Bridge of Instability.
4:45	This may look easy, but the trick is it attaches at a single point on each end.	This may look easy, but the trick is it attaches at a single point on each end.
4:51	From a physics standpoint,	From a physics standpoint,
4:52	that makes it no different than trying to crawl across a tightrope.	that makes it no different than trying to crawl across a tightrope.
4:55	It's the same deceptive principle with that carnival game that's impossible.	It's the same deceptive principle with that carnival game that's impossible.
4:59	In another video, I used physics to show which carnival games were basically scams	In another video, I used physics to show which carnival games were basically scams
5:04	and how to beat them.	and how to beat them.
5:05	Next up, the Maze of a Thousand Corridors. This one's a bit more cerebral.	Next up, the Maze of a Thousand Corridors. This one's a bit more cerebral.
5:09	If they make it through, it'll bring them to the Pitchfork Tumblers,	If they make it through, it'll bring them to the Pitchfork Tumblers,
5:12	inspired by the show Wipeout.	inspired by the show Wipeout.

5:14	Unlike Wipeout, I should point out that all these contraptions	Unlike Wipeout, I should point out that all these contraptions
5:17	are squirrel-friendly.	are squirrel-friendly.
5:19	And even with a little back force, they will break away and stop spinning.	And even with a little back force, they will break away and stop spinning.
5:22	If they make it past the Pitchfork Tumblers of Treachery,	If they make it past the Pitchfork Tumblers of Treachery,
5:25	they come to one of my favorites, the Homewrecker.	they come to one of my favorites, the Homewrecker.
5:35	I put her in to tempt the squirrel to lose focus on the prize at the end.	I put her in to tempt the squirrel to lose focus on the prize at the end.
5:39	This pad has a pressure sensor underneath, and it connects to a microcontroller	This pad has a pressure sensor underneath, and it connects to a microcontroller
5:43	that has a relay that connects to a solenoid	that has a relay that connects to a solenoid
5:45	that connects to a pneumatic piston that connects to this pressure sensor.	that connects to a pneumatic piston that connects to this pressure sensor.
5:49	If they stand here for more than three seconds...	If they stand here for more than three seconds...
5:53	...they've gotta start all over again.	...they've gotta start all over again.
5:55	Next up is the Slinky Bridge of Deception.	Next up is the Slinky Bridge of Deception.
5:58	I've suspended a juicy walnut right in the middle.	I've suspended a juicy walnut right in the middle.
6:01	No way this bridge will support their weight,	No way this bridge will support their weight,
6:03	so it'll be interesting how they approach it.	so it'll be interesting how they approach it.
6:05	The sixth challenge is the Tourist Trap. It's a reward for making it this far.	The sixth challenge is the Tourist Trap. It's a reward for making it this far.
6:10	If the squirrel sticks their head through this board,	If the squirrel sticks their head through this board,
6:12	it's a photo op they can hang on the wall of their nest.	it's a photo op they can hang on the wall of their nest.
6:15	I'll place a walnut here as an incentive for them not to miss this opportunity.	I'll place a walnut here as an incentive for them not to miss this opportunity.
6:19	The penultimate challenge are the Quad Steps of Great Elevation.	The penultimate challenge are the Quad Steps of Great Elevation.
6:22	Straight from Ninja Warrior, this is super tricky	Straight from Ninja Warrior, this is super tricky
6:25	because these pads are all totally smooth and placed at a 45-degree incline.	because these pads are all totally smooth and placed at a 45-degree incline.
6:29	Now the last challenge.	Now the last challenge.
6:30	At this point, they are a mere few feet away from their ultimate goal.	At this point, they are a mere few feet away from their ultimate goal.
6:34	If they're not careful, they'll have to deal with the Orbital-Assist Platform,	If they're not careful, they'll have to deal with the Orbital-Assist Platform,

6:38	aka the Final Countdown,	aka the Final Countdown,
6:40	aka it's not a catapult, it's a squirrel-apult.	aka it's not a catapult, it's a squirrel-apult.
6:46	Which means they now have to go back and start all the way from the beginning.	Which means they now have to go back and start all the way from the beginning.
6:50	If they wanna avoid that fate,	If they wanna avoid that fate,
6:52	they just need to stand on this pad for less than three seconds.	they just need to stand on this pad for less than three seconds.
6:55	But once again, I've placed a diversion here.	But once again, I've placed a diversion here.
6:57	This projects an optical illusion of anything you place in this dish.	This projects an optical illusion of anything you place in this dish.
7:01	It's super trippy, and looks even more impressive in real life	It's super trippy, and looks even more impressive in real life
7:04	because the projected image is 3D.	because the projected image is 3D.
7:06	They're nine bucks on Amazon, and I put a link	They're nine bucks on Amazon, and I put a link
7:08	that explains the juicy physics of how they work in the video description.	that explains the juicy physics of how they work in the video description.
7:11	With regards to squirrel safety, I know I seem to have this reputation of...	With regards to squirrel safety, I know I seem to have this reputation of...
7:22	...improvised high-speed projectiles,	...improvised high-speed projectiles,
7:23	so this one might worry a few animal lovers,	so this one might worry a few animal lovers,
7:26	including my wife.	including my wife.
7:27	We have five pets and one child. The child was the only one I had a say in.	We have five pets and one child. The child was the only one I had a say in.
7:31	So rest assured, I can adjust the pressure on the piston,	So rest assured, I can adjust the pressure on the piston,
7:35	and we'll be staying way below any launch force	and we'll be staying way below any launch force
7:37	that could hurt any of our adorable backyard contestants.	that could hurt any of our adorable backyard contestants.
7:40	Finally, after making it through all that,	Finally, after making it through all that,
7:42	we're back at the world's most satisfying bird feeder.	we're back at the world's most satisfying bird feeder.
7:45	As far as I'm concerned, if they make it here, they've earned the right	As far as I'm concerned, if they make it here, they've earned the right
7:49	to gorge next to the birds for as long as they want.	to gorge next to the birds for as long as they want.
7:51	After nearly a month of prep work,	After nearly a month of prep work,
7:54	the course was officially open for business.	the course was officially open for business.
7:58	They could smell the walnuts at the final station.	They could smell the walnuts at the final station.
8:00	You could see the gears turning	You could see the gears turning

8:02	because right above that was their favorite bird feeder.	because right above that was their favorite bird feeder.
8:06	And so before long, we had our first contender.	And so before long, we had our first contender.
8:26	That didn't exactly go as planned, so Rick decides to regroup.	That didn't exactly go as planned, so Rick decides to regroup.
8:29	Speaking of which, that's Rick.	Speaking of which, that's Rick.
8:31	There are a total of four competitors attempting the course.	There are a total of four competitors attempting the course.
8:35	First up is Rick, and he weighs 500 grams.	First up is Rick, and he weighs 500 grams.
8:38	I know this 'cause he's standing on a custom-made scale.	I know this 'cause he's standing on a custom-made scale.
8:41	Rick is very clever, as you'll see in a minute,	Rick is very clever, as you'll see in a minute,
8:43	but he gets spooked easily.	but he gets spooked easily.
8:45	Next up is Marty. He's basically indistinguishable from Rick.	Next up is Marty. He's basically indistinguishable from Rick.
8:48	Rick and Marty are inseparable and spend all their time together.	Rick and Marty are inseparable and spend all their time together.
8:52	Our third contender is Frank, also coming in around 500 grams.	Our third contender is Frank, also coming in around 500 grams.
8:56	He's very gutsy. He's also kinda dumb.	He's very gutsy. He's also kinda dumb.
8:58	Finally, we've got my personal favorite, Phat Gus.	Finally, we've got my personal favorite, Phat Gus.
9:02	He's just really charming and will pose if he sees a camera.	He's just really charming and will pose if he sees a camera.
9:05	He also really likes to eat,	He also really likes to eat,
9:07	and he tips the scales at an impressive 800 grams.	and he tips the scales at an impressive 800 grams.
9:10	While most of the squirrels are particular	While most of the squirrels are particular
9:12	before making a selection and then gracefully leaping away,	before making a selection and then gracefully leaping away,
9:15	Phat Gus just goes for it, hands-free.	Phat Gus just goes for it, hands-free.
9:18	Just makes himself real comfortable.	Just makes himself real comfortable.
9:20	Here he is laying out with a light snack.	Here he is laying out with a light snack.
9:23	You can't fault the guy for just clearly appreciating a good meal.	You can't fault the guy for just clearly appreciating a good meal.
9:27	So now that you know our four contenders, let's get back to Rick.	So now that you know our four contenders, let's get back to Rick.
9:30	He hops back on, and he's doing better than his first attempt.	He hops back on, and he's doing better than his first attempt.
9:34	Mm, just barely.	Mm, just barely.
9:36	This is the best because he's like, "Okay, I got this."	This is the best because he's like, "Okay, I got this."

9:40	"Never mind. Well, maybe... No, I definitely don't."	"Never mind. Well, maybe... No, I definitely don't."
9:43	This was the exact moment I realized I just might be outmatched here.	This was the exact moment I realized I just might be outmatched here.
9:53	Marty was watching Rick, and after a dip of his toes in the water,	Marty was watching Rick, and after a dip of his toes in the water,
9:56	adopts his strategy right out of the gate.	adopts his strategy right out of the gate.
9:58	Oof. So close.	Oof. So close.
10:00	But an excellent recovery.	But an excellent recovery.
10:01	And Phat Gus, perhaps predictably, comes up a little short,	And Phat Gus, perhaps predictably, comes up a little short,
10:05	which brings us to the Maze of a Thousand Corridors,	which brings us to the Maze of a Thousand Corridors,
10:07	which, to be honest, they mastered this one pretty quickly.	which, to be honest, they mastered this one pretty quickly.
10:12	Here's a solid run from Phat Gus.	Here's a solid run from Phat Gus.
10:16	And now for the Pitchfork Tumblers of Treachery.	And now for the Pitchfork Tumblers of Treachery.
10:19	Marty was like, "Yeah, that's a hard pass."	Marty was like, "Yeah, that's a hard pass."
10:22	Phat Gus is just like, "Abort!"	Phat Gus is just like, "Abort!"
10:25	Frank is just like, "Nope."	Frank is just like, "Nope."
10:27	Then he's like, "Wait a second. Hmm. Still nope."	Then he's like, "Wait a second. Hmm. Still nope."
10:30	Then Marty flies in real hot over the top.	Then Marty flies in real hot over the top.
10:33	That's partial credit as far as I'm concerned.	That's partial credit as far as I'm concerned.
10:35	And then here's Phat Gus with my favorite attempt by far.	And then here's Phat Gus with my favorite attempt by far.
10:44	Frank shows no fear, which helps him pull off	Frank shows no fear, which helps him pull off
10:47	this near flawless run.	this near flawless run.
10:48	Finally, Rick weaves his way through, essentially untouched.	Finally, Rick weaves his way through, essentially untouched.
10:52	The squirrels were most active from 6:00 to 10:00 a.m.,	The squirrels were most active from 6:00 to 10:00 a.m.,
10:55	so I've been getting up before sunrise to start filming from in my house	so I've been getting up before sunrise to start filming from in my house
11:00	like one of those camouflage nature photographers,	like one of those camouflage nature photographers,
11:02	only, like, a suburbanized version.	only, like, a suburbanized version.
11:05	I've also had a handful of streaming cameras rolling 24-7,	I've also had a handful of streaming cameras rolling 24-7,
11:07	which has allowed me to make such discoveries as	which has allowed me to make such discoveries as
11:10	we have the world's worst guard dog.	we have the world's worst guard dog.

11:12	Thanks for the help, pal.	Thanks for the help, pal.
11:13	Now back to the Homewrecker.	Now back to the Homewrecker.
11:15	Rick is instinctively suspicious of this gold digger,	Rick is instinctively suspicious of this gold digger,
11:18	and only for a moment puts his full weight on the platform.	and only for a moment puts his full weight on the platform.
11:21	I told you he was smart. Marty, on the other hand...	I told you he was smart. Marty, on the other hand...
11:31	She is a cruel mistress.	She is a cruel mistress.
11:34	Because I know my wife is going to ask,	Because I know my wife is going to ask,
11:36	no, this will not hurt the squirrel for two reasons.	no, this will not hurt the squirrel for two reasons.
11:39	Number one, they are amazing, like cats, and always land on their feet.	Number one, they are amazing, like cats, and always land on their feet.
11:43	And two, they themselves were jumping from this height all the time,	And two, they themselves were jumping from this height all the time,
11:47	and in some cases, much higher.	and in some cases, much higher.
11:49	Squirrels are one of the few mammals that can survive a fall from any height	Squirrels are one of the few mammals that can survive a fall from any height
11:53	because they make their terminal velocity so low.	because they make their terminal velocity so low.
11:56	You can see it here.	You can see it here.
11:57	They make their body as flat as possible, then put their tail back to increase drag.	They make their body as flat as possible, then put their tail back to increase drag.
12:01	They basically become their own parachute.	They basically become their own parachute.
12:03	Before they land, they pull in and fully extend their arms and legs	Before they land, they pull in and fully extend their arms and legs
12:07	to act as shock-absorbing landing gear.	to act as shock-absorbing landing gear.
12:09	This is so cool.	This is so cool.
12:11	Back to the course, where even smart Rick lets his guard down,	Back to the course, where even smart Rick lets his guard down,
12:15	which is exactly when she strikes.	which is exactly when she strikes.
12:20	But like all the other obstacles so far, after a day or two, they'd mastered it.	But like all the other obstacles so far, after a day or two, they'd mastered it.
12:25	In this case, even disregarding her offering of a fresh walnut.	In this case, even disregarding her offering of a fresh walnut.
12:30	Now onto the Slinky Bridge of Deception. This was the sleeper obstacle in my book.	Now onto the Slinky Bridge of Deception. This was the sleeper obstacle in my book.
12:34	That is not a large gap, which they can easily jump,	That is not a large gap, which they can easily jump,
12:37	but having the Slinky there threw them off somehow.	but having the Slinky there threw them off somehow.
12:40	You can see how much Frank equivocates here, and he's the brave one.	You can see how much Frank equivocates here, and he's the brave one.

12:45	And Rick only jumps when Marty shows up for emotional support.	And Rick only jumps when Marty shows up for emotional support.
12:50	But like all the obstacles so far,	But like all the obstacles so far,
12:52	after a day or two, they'd mastered it.	after a day or two, they'd mastered it.
12:54	Now onto our third-to-last obstacle, the Tourist Trap.	Now onto our third-to-last obstacle, the Tourist Trap.
12:57	This was just a freaking delight.	This was just a freaking delight.
12:59	A little peanut butter really did the trick here.	A little peanut butter really did the trick here.
13:13	Now here comes Phat Gus, and how Phat Gus is this?	Now here comes Phat Gus, and how Phat Gus is this?
13:16	He sees the camera and he's like, "Paint me like one of your French squirrels."	He sees the camera and he's like, "Paint me like one of your French squirrels."
13:20	Yeah? No? No takers? All right.	Yeah? No? No takers? All right.
13:24	Here's another classic Gus, where he looks at the camera...	Here's another classic Gus, where he looks at the camera...
13:27	Wait a second. Play that back.	Wait a second. Play that back.
13:30	And freeze. Zoom.	And freeze. Zoom.
13:32	Enhance.	Enhance.
13:43	Well, turns out Phat Gus is not a dude.	Well, turns out Phat Gus is not a dude.
13:45	And he's pregnant.	And he's pregnant.
13:47	I was suddenly feeling real uncomfortable about all those weight comments.	I was suddenly feeling real uncomfortable about all those weight comments.
13:51	So after a bit of a pivot to smooth things over,	So after a bit of a pivot to smooth things over,
13:54	we soldiered on to the Quad Steps of Great Elevation.	we soldiered on to the Quad Steps of Great Elevation.
13:57	Frank comes in rather confident, almost to his demise,	Frank comes in rather confident, almost to his demise,
14:01	but he pulls off the smooth recovery.	but he pulls off the smooth recovery.
14:07	After not much time, they were looking pretty good here.	After not much time, they were looking pretty good here.
14:10	In fact, one morning, I noticed Marty skipped the last step altogether,	In fact, one morning, I noticed Marty skipped the last step altogether,
14:14	which I took as a challenge.	which I took as a challenge.
14:16	I removed the middle step to see what they would do.	I removed the middle step to see what they would do.
14:18	And who else but Phantastic Gus?	And who else but Phantastic Gus?
14:21	Let's slow that down a bit.	Let's slow that down a bit.
14:24	They're such amazing jumpers,	They're such amazing jumpers,
14:26	so I suppose the quick mastery here wasn't a surprise.	so I suppose the quick mastery here wasn't a surprise.
14:29	This right here is probably a ten-foot jump.	This right here is probably a ten-foot jump.

14:31	And here's some classic brave-but-dumb Frank.	And here's some classic brave-but-dumb Frank.
14:34	He'd already made it to the Homewrecker at least a dozen times by this point.	He'd already made it to the Homewrecker at least a dozen times by this point.
14:38	And yet he decides this 12-foot jump is his best course of action.	And yet he decides this 12-foot jump is his best course of action.
14:43	Smart Rick was the first to enter the course,	Smart Rick was the first to enter the course,
14:45	so in the end, after about a week of effort,	so in the end, after about a week of effort,
14:47	it was fitting he was the one to make it to the jackpot first.	it was fitting he was the one to make it to the jackpot first.
14:50	He's so stoked, he wants to run the whole course in reverse.	He's so stoked, he wants to run the whole course in reverse.
14:54	In hindsight, we should have made the jackpot a little quieter.	In hindsight, we should have made the jackpot a little quieter.
14:57	Here's Marty, as always, not far behind.	Here's Marty, as always, not far behind.
15:05	He's also a bit flighty, which worked against him here.	He's also a bit flighty, which worked against him here.
15:08	Down below, Phantastic Gus, ever the opportunist,	Down below, Phantastic Gus, ever the opportunist,
15:11	comes through with a real stroke of brilliance.	comes through with a real stroke of brilliance.
15:14	But pretty soon, Marty comes back like, "Um, I'm the one who kinda earned that."	But pretty soon, Marty comes back like, "Um, I'm the one who kinda earned that."
15:19	And then Phantastic Gus is like, "You snooze, you lose, sucker!"	And then Phantastic Gus is like, "You snooze, you lose, sucker!"
15:22	Might I say, Phantastic Gus, when you sit like that,	Might I say, Phantastic Gus, when you sit like that,
15:25	you don't look an ounce over 700 grams.	you don't look an ounce over 700 grams.
15:27	They learned that sound was good and quickly returned for their reward	They learned that sound was good and quickly returned for their reward
15:31	before Phantastic Gus got any ideas.	before Phantastic Gus got any ideas.
15:33	Here's a single camera shot of a full run,	Here's a single camera shot of a full run,
15:35	By the end, all of them could do the full course in less than 40 seconds.	By the end, all of them could do the full course in less than 40 seconds.
15:39	And you might be wondering what happened to the squirrel-apult.	And you might be wondering what happened to the squirrel-apult.
15:42	Unfortunately, they didn't fall for my illusion much,	Unfortunately, they didn't fall for my illusion much,
15:46	but I'm about to show you the two times they did.	but I'm about to show you the two times they did.
15:48	As a promise to my wife, I turned the pressure to only 40% full power,	As a promise to my wife, I turned the pressure to only 40% full power,
15:52	so the acceleration the squirrel would feel is about half a G,	so the acceleration the squirrel would feel is about half a G,

15:55	which is ten times less than a typical roller coaster.	which is ten times less than a typical roller coaster.
15:57	Additionally, I never turned on the Homewrecker trapdoor	Additionally, I never turned on the Homewrecker trapdoor
16:00	or the launch platform	or the launch platform
16:01	when I wasn't actively watching from inside my house,	when I wasn't actively watching from inside my house,
16:04	where I had a manual-override button.	where I had a manual-override button.
16:06	Because something about launching Phantastic Gus just didn't feel right.	Because something about launching Phantastic Gus just didn't feel right.
16:10	What you're about to see is gonna look chaotic, but when we check the slo-mo,	What you're about to see is gonna look chaotic, but when we check the slo-mo,
16:14	amazingly, I'll show you	amazingly, I'll show you
16:15	how the squirrel is totally in control pretty much the whole time.	how the squirrel is totally in control pretty much the whole time.
16:22	Here's a second angle.	Here's a second angle.
16:26	Even if you don't trust my calculations,	Even if you don't trust my calculations,
16:28	I can assure you that the squirrels did not care.	I can assure you that the squirrels did not care.
16:31	In both launch cases, they were back up on the course in five minutes.	In both launch cases, they were back up on the course in five minutes.
16:35	Here's the slo-mo, and see if you can spot the two critical things he does	Here's the slo-mo, and see if you can spot the two critical things he does
16:38	to make sure he lands feet down.	to make sure he lands feet down.
16:41	Did you catch it?	Did you catch it?
16:44	The first critical moment is here.	The first critical moment is here.
16:46	He's already done the math	He's already done the math
16:48	and he's locked his head on the spot where he plans to land,	and he's locked his head on the spot where he plans to land,
16:51	and now his head will lock on that spot the whole way down.	and now his head will lock on that spot the whole way down.
16:55	What's crazy is the time it takes him to do all the predictive math	What's crazy is the time it takes him to do all the predictive math
16:58	to know where he will land and to lock his head	to know where he will land and to lock his head
17:00	is 300 milliseconds.	is 300 milliseconds.
17:02	That's exactly this long. Less than the blink of an eye.	That's exactly this long. Less than the blink of an eye.
17:05	That blows my mind, but it makes sense these reactions evolved to be razor sharp	That blows my mind, but it makes sense these reactions evolved to be razor sharp
17:10	if you live in the trees, constantly making split-second decisions.	if you live in the trees, constantly making split-second decisions.
17:14	Besides picking the landing spot with his head,	Besides picking the landing spot with his head,
17:16	the second trick is with his arms.	the second trick is with his arms.

17:18	To keep his upper body from rotating, he has his arms out.	To keep his upper body from rotating, he has his arms out.
17:21	The moment he wants to rotate his upper body quickly, he tucks them in.	The moment he wants to rotate his upper body quickly, he tucks them in.
17:25	When he wants to stop rotating, he extends them out again.	When he wants to stop rotating, he extends them out again.
17:28	This is conservation of angular momentum.	This is conservation of angular momentum.
17:31	It's the same trick ice-skaters use.	It's the same trick ice-skaters use.
17:33	Bring your limbs in if you want to rotate fast.	Bring your limbs in if you want to rotate fast.
17:35	Extend them out if you want to slow the rotation.	Extend them out if you want to slow the rotation.
17:38	My buddy Destin from Smarter Every Day also made a video	My buddy Destin from Smarter Every Day also made a video
17:40	analyzing how cats always land on their feet,	analyzing how cats always land on their feet,
17:43	and they do the exact same two tricks.	and they do the exact same two tricks.
17:45	The head target-lock, and bringing limbs in to rotate	The head target-lock, and bringing limbs in to rotate
17:48	and out to stop rotation.	and out to stop rotation.
17:49	Here's the second launch.	Here's the second launch.
17:52	This may look like pure luck in real time, but the slo-mo tells a different story.	This may look like pure luck in real time, but the slo-mo tells a different story.
17:56	In that amount of time, he's already made a slight lateral push	In that amount of time, he's already made a slight lateral push
18:00	and locked his landing target with his head.	and locked his landing target with his head.
18:02	We know from the previous clip, he can rotate his body to any orientation.	We know from the previous clip, he can rotate his body to any orientation.
18:06	He doesn't in this case because he sees his small push	He doesn't in this case because he sees his small push
18:09	has already aligned him for the perfect landing.	has already aligned him for the perfect landing.
18:11	That's just so cool.	That's just so cool.
18:12	After those two physics demonstrations,	After those two physics demonstrations,
18:14	I disabled both the launcher and the Homewrecker trapdoor	I disabled both the launcher and the Homewrecker trapdoor
18:17	and let them enjoy the course in peace.	and let them enjoy the course in peace.
18:19	They had earned it.	They had earned it.
18:21	Plus, after spending a month together, we were sort of bros.	Plus, after spending a month together, we were sort of bros.
18:24	If they saw me coming out, they would jump away,	If they saw me coming out, they would jump away,
18:27	but they'd quickly be right back.	but they'd quickly be right back.

18:28	Sadly, all good things must come to an end,	Sadly, all good things must come to an end,
18:31	and because I didn't want them to develop a dependence on my handouts,	and because I didn't want them to develop a dependence on my handouts,
18:34	after a week of free rein, I took down the course.	after a week of free rein, I took down the course.
18:37	But in its place, I put up a specialized squirrel-sized picnic table.	But in its place, I put up a specialized squirrel-sized picnic table.
18:40	Every morning, this door opens to release a few walnuts	Every morning, this door opens to release a few walnuts
18:43	as a treat to remind them of our good times together.	as a treat to remind them of our good times together.
18:46	Although I had to laugh that somehow I didn't anticipate, on day one,	Although I had to laugh that somehow I didn't anticipate, on day one,
18:50	Frank would try and figure out	Frank would try and figure out
18:52	and partially succeed in finding a way to beat the system.	and partially succeed in finding a way to beat the system.
18:55	Like everything, they quickly caught on,	Like everything, they quickly caught on,
18:57	and on some mornings, I'll catch them waiting for the daily ration,	and on some mornings, I'll catch them waiting for the daily ration,
19:00	chilling patiently on the bench seat.	chilling patiently on the bench seat.
19:02	That is, everyone but	That is, everyone but
19:04	Phantastic Gus.	Phantastic Gus.
19:05	Don't ever change, buddy.	Don't ever change, buddy.
19:07	When I started this two-month-long adventure, trying to bird-watch,	When I started this two-month-long adventure, trying to bird-watch,
19:10	I had no idea I would enjoy squirrel-watching so much more.	I had no idea I would enjoy squirrel-watching so much more.
19:14	It's made me realize	It's made me realize
19:15	that, even amongst the structures and pavement and power lines,	that, even amongst the structures and pavement and power lines,
19:19	how interesting nature can be in a single suburban backyard	how interesting nature can be in a single suburban backyard
19:22	if you just really stop to look.	if you just really stop to look.
19:24	Even the plants are amazing.	Even the plants are amazing.
19:25	I had no idea that as vines grow,	I had no idea that as vines grow,
19:27	they spin around, looking for something to grab onto.	they spin around, looking for something to grab onto.
19:30	I didn't anticipate the wide variety of animals	I didn't anticipate the wide variety of animals
19:32	that use fences as inter-yard highways to get around	that use fences as inter-yard highways to get around
19:36	as a substitute for all the trees that existed before we got here.	as a substitute for all the trees that existed before we got here.

19:39	Even little interactions, like this spider extending his life	Even little interactions, like this spider extending his life
19:43	with this very timely leap into the unknown,	with this very timely leap into the unknown,
19:45	were just so delightful to see.	were just so delightful to see.
19:47	But of all the animals, I can definitively say	But of all the animals, I can definitively say
19:49	that squirrels are my absolute favorite,	that squirrels are my absolute favorite,
19:52	and they're a more formidable adversary than porch pirates.	and they're a more formidable adversary than porch pirates.
19:55	They're kind of adorable, incredibly crafty,	They're kind of adorable, incredibly crafty,
19:58	curious by nature, athletic, and persistent.	curious by nature, athletic, and persistent.
20:01	Turns out squirrels can live to be 20 years old.	Turns out squirrels can live to be 20 years old.
20:04	I like to think that, someday,	I like to think that, someday,
20:05	Phantastic Gus will bring his grand- squirrels to the fence	Phantastic Gus will bring his grand- squirrels to the fence
20:08	and regale them with tales of cowboys, and courage, and legendary walnut piles.	and regale them with tales of cowboys, and courage, and legendary walnut piles.